Incremental Private Set Intersection Cardinality

Abstract

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A M's zk proof

A.1 I-PSI-CA-F

Recall in Sec. **??**, we want to prove that M knows $\mathsf{tsk}_2, \mathsf{y}, r, \{\mathsf{e_d}\}_{d=0}^{n}$ such that

$$\vec{\mathsf{y}} = \mathcal{F}(\mathsf{y}) \wedge \tag{1}$$

$$\vec{c_{\mathsf{M}}} = \mathsf{Enc}_{\mathsf{pk}_2}(\vec{\mathsf{y}}) \wedge \tag{2}$$

$$c_{\vec{a}} = \mathsf{Dec}_{\mathsf{tsk}_2}(E) \wedge \mathsf{tpk} = {\mathsf{tpk}_1}^{\mathsf{tsk}_2} \wedge \tag{3}$$

$$E = \sum_{d=0}^{n}{}_{\mathsf{tpk}} \ \mathsf{e_d} \times_{\mathsf{tpk}} \vec{a}[d] \wedge \tag{4}$$

$$\forall d \in [0, n]\,,$$

$$\mathsf{e_d} = r \cdot \mathsf{y}^d \tag{5}$$

$$\exists\, \delta_0, \ldots, \delta_{\mathsf{n}} \in \mathbb{F}^* :$$

$$\vec{p}[d] = \vec{a}[d] \times_{\mathsf{tpk}} \mathsf{y} +_{\mathsf{tpk}} \mathsf{Enc}_{\mathsf{tpk}}(0; \delta_{\mathsf{d}})\} \tag{6}$$

with public parameters $c_{\mathsf{M}}, E, \mathsf{tpk}, \mathsf{tpk}_1, \vec{c}_{\mathsf{T}}, \vec{a}$.

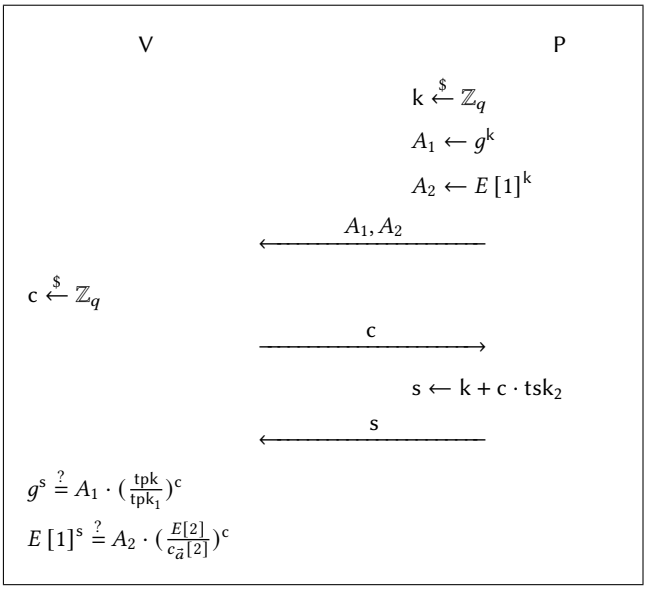


Figure 1: Proof π_{dec} for (3): $c_{\vec{a}} = \mathsf{Dec}_{\mathsf{tsk}_2}(E) \wedge \mathsf{tpk} = {\mathsf{tpk}_1}^{\mathsf{tsk}_2}$

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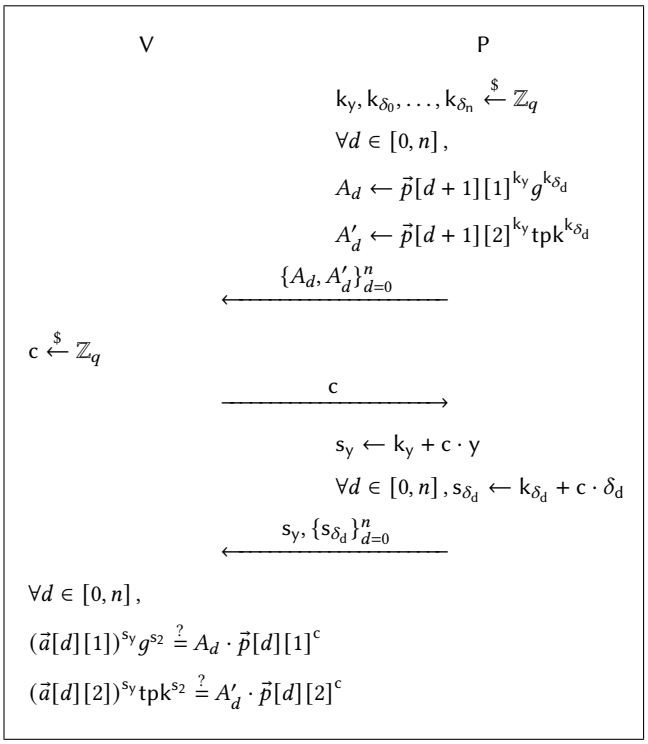


Figure 2: Proof π_{next} for (6): $\vec{p}[d] = \vec{a}[d] \times_{\mathsf{tpk}} \mathsf{y} +_{\mathsf{tpk}} \mathsf{Enc}_{\mathsf{tpk}}(0; \delta_\mathsf{d})$

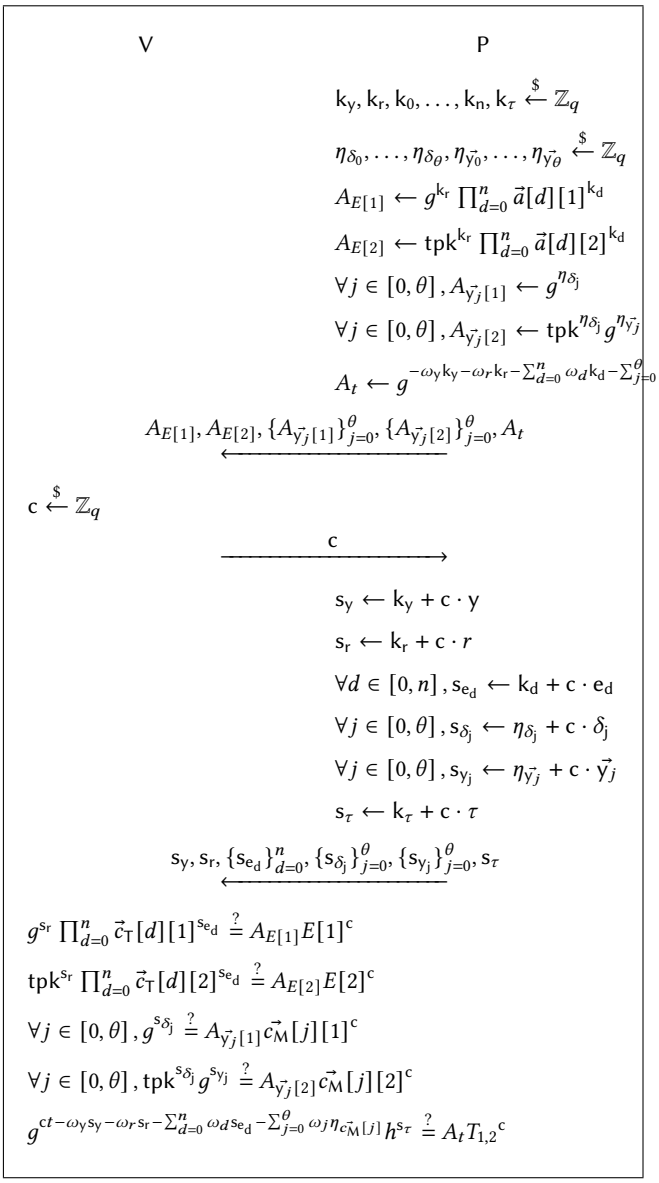


Figure 3: Proof π_Σ for Σ-Bullets Conjunction of (1), (2), (4), (5).